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Letter

Formal Total Synthesis of (−)-Isolaurallene and Structure Determination of (+)-4-*epi*-Isolaurallene via Stereodivergent Intramolecular Amide Enolate Alkylation

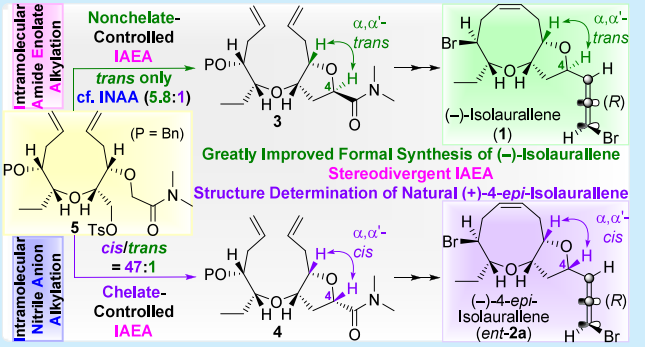
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ABSTRACT: Described herein are the formal total syntheses of (−)-isolaurallene (**1**) and structure determination of newly isolated (+)-4-*epi*-isolaurallene (**2**) via a highly stereodivergent intramolecular amide enolate alkylation (IAEA) strategy. Highly stereoselective constructions of *α,α′-trans*-THF **3** and *α,α′-cis*-THF **4** were accomplished by subjecting common intermediate **5** to non-chelate-controlled and chelate-controlled IAEA, respectively. The former protocol greatly improves the moderate stereoselectivity previously observed in constructing *α,α′-trans*-THF rings via intramolecular nitrile anion alkylation (INAA).



The *Laurencia* genus of red algae has yielded a plethora of oxacyclic halogenated C_{15} acetogenins, some of which feature a stereoisomeric dioxabicyclic bromoallene skeleton comprising the 2,10-dioxabicyclo[7.3.0]dodecene moiety.[1] Among these, the first natural product with this structure was (−)-isolaurallene (**1**), which was isolated from the red alga *Laurencia nipponica* by Kurata and co-workers in 1982.[2] As shown in Figure 1, spectroscopic and single-crystal X-ray crystallographic studies revealed that (−)-isolaurallene (**1**) is composed of an *α,α′-cis*-oxonene unit fused with an *α,α′-trans*-THF ring, which is substituted with an (*R*)-bromoallene appendage at C(4). The first total synthesis of this natural product was accomplished by the Crimmins group, who employed an Evans asymmetric allylation and nine-membered ether ring-closing metathesis (RCM) to the *α,α′-cis*-oxonene as key steps.[3] The *α,α′-trans*-THF ring in **1** was then constructed by an asymmetric epoxidation/internal Williamson etherification sequence. Subsequently, our group achieved a total synthesis of **1** utilizing a stereoselective intermolecular amide enolate alkylation/RCM strategy for the *α,α′-cis*-oxonene and intramolecular nitrile anion alkylation (INAA) for the *α,α′-trans*-THF ring.[4]

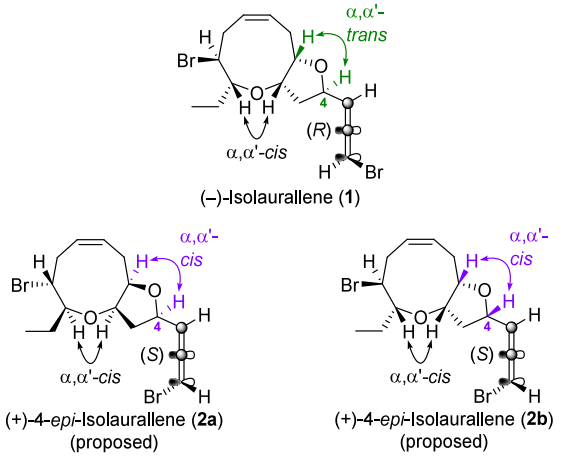


Figure 1. (−)-Isolaurallene (**1**) and two proposed structures of (+)-4-*epi*-isolaurallene (**2**).

Meanwhile, (+)-4-*epi*-isolaurallene (**2**), another closely related C_{15} acetogenin with a 2,10-dioxabicyclo[7.3.0]-dodecene unit, was recently isolated from a population of marine red alga *Laurencia nangii* Masuda by Vairappan and co-workers in 2020.[5] The structure of (+)-4-*epi*-isolaurallene was elucidated from extensive spectroscopic studies. A comparison of its ^{1}H NMR spectrum with those of related natural and unnatural compounds revealed that the 2,10-dioxabicyclo[7.3.0]dodecene unit comprises an *α,α′-cis*-oxo-

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nene unit and an α,α′-*cis*-THF ring. Judging from the strongly positive rotation of (+)-4-*epi*-isolaurallene ($[\alpha]^{25}_{D}$ +56.1 (*c* 0.46, $CHCl_3$)), the absolute stereochemistry of the bromoallene moiety was assigned as *S* on the basis of Lowe's rule.[6] However, since the relative stereochemistry between the dioxabicyclic core and the bromoallene moiety could not be determined using spectroscopic methods, the structure of (+)-4-*epi*-isolaurallene was proposed to be either **2a** or **2b** (Figure 1). No total synthesis of this natural product has been reported to date, and thus, its structure has remained unconfirmed.

As stated above, we recently reported the stereoselective substrate-controlled total synthesis of (−)-isolaurallene (**1**), which involved the construction of the α,α′-*trans*-THF moiety via INAA with a *trans*:*cis* selectivity of 5.8:1. In an effort to improve this modest stereoselectivity, and based on the insights acquired during our recent total syntheses of other tetrahydrofuranoid natural products, we envisioned that the use of KHMDS as the base in the presence of a crown ether might produce an α,α′-*trans*-THF in a highly stereoselective manner, thereby enabling a stereodivergent intramolecular amide enolate alkylation (IAEA) strategy for the selective construction of both α,α′-*cis* and α,α′-*trans* THF moieties from a common substrate.[7a]

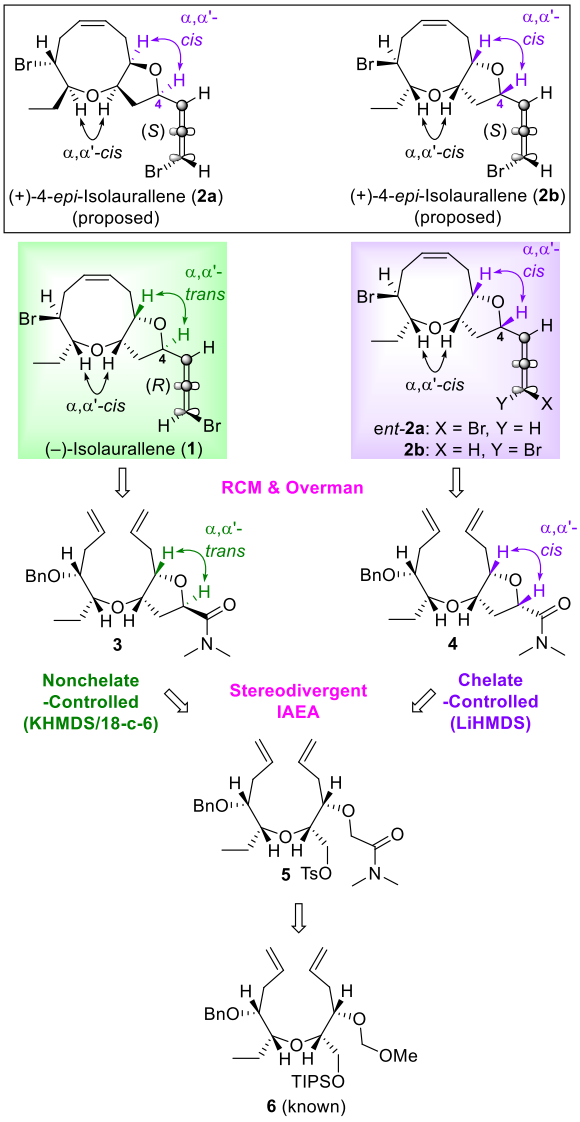
Herein, we describe the formal total synthesis of (−)-isolaurallene (**1**) and consequent structure determination of (+)-4-*epi*-isolaurallene (**2**) via the highly stereoselective total syntheses of *ent*-**2a** and **2b** employing our highly stereodivergent IAEA strategy in a substrate-controlled manner.

As shown in our retrosynthetic plan (Scheme 1), we were confident that α,α′-*trans*-THF amide **3** and its C(4) epimer, α,α′-*cis*-THF amide **4**, could lead to (−)-isolaurallene (**1**) and 4-*epi*-isolaurallenes (*ent*-**2a** and **2b**), respectively, via the RCM and Overman protocol. It is worth mentioning at this point that we opted to synthesize *ent*-**2a** and **2b** instead of the originally proposed structures **2a** and **2b** for the sake of simplicity in our approach to determine the relative stereochemistry between the dioxabicyclic core and the bromoallene appendage of (+)-4-*epi*-isolaurallene (**2**). Moreover, with the knowledge accrued during the aforementioned recent total syntheses of natural products containing a THF ring,[7] we envisaged that pivotal intermediates α,α′-*trans*-THF **3** and α,α′-*cis*-THF **4** could be constructed via non-chelate-controlled and chelate-controlled IAEA, respectively, in a stereodivergent fashion from common substrate ω-tosyl α-alkoxyamide **5**, which could be obtained from known compound **6**.[4]

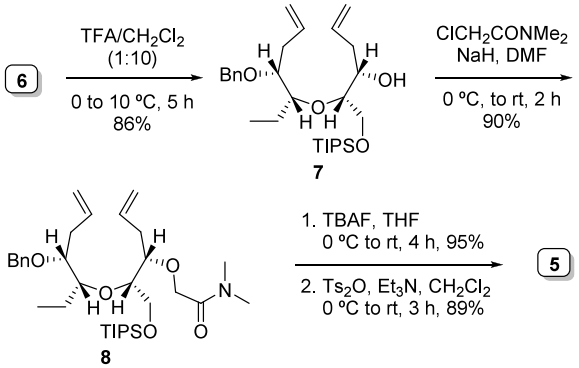
Our synthesis commenced with the preparation of stereodivergent IAEA substrate **5** as described in Scheme 2. Known compound **6** was converted into IAEA substrate ω-tosyl α-alkoxyamide **5** in a good overall yield (66%) via a straightforward four-step sequence. Thus, chemoselective deprotection of the MOM group in the presence of the TIPS group in **6** by treatment with trifluoroacetic acid and subsequent Williamson etherification of the resulting secondary alcohol **7** with *N*,*N*-dimethyl chloroacetamide furnished α-alkoxyamide **8**. Removal of the TIPS protecting group in **8** with TBAF followed by tosylation of the resultant primary alcohol with *p*-toluenesulfonic anhydride delivered the desired internal alkylation substrate **5**.

With ω-tosyl α-alkoxyamide **5** in hand, we proceeded to implement the pivotal stereodivergent IAEA reaction in the construction of both α,α′-*trans*-THF **3** and α,α′-*cis*-THF **4**, as depicted in Scheme 3. As anticipated, treatment of **5** with LiHMDS in toluene at −78 to −50 °C for 3 h furnished the desired α,α′-*cis*-THF amide **4** in excellent yield (93%) and diastereoselectivity (47:1 *cis*:*trans*, by ^{1}H NMR analysis),

Scheme 1. Retrosynthetic Plan for (−)-Isolaurallene (1) and *ent*-2a and 2b

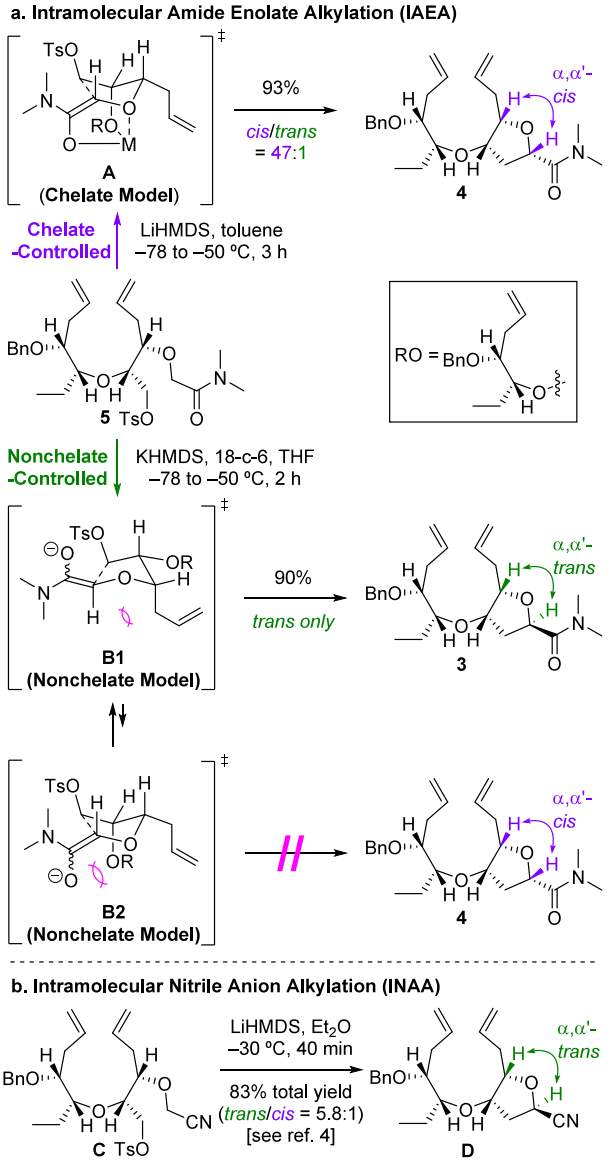


Scheme 2. Preparation of Stereodivergent Intramolecular Amide Enolate Alkylation Substrate 5



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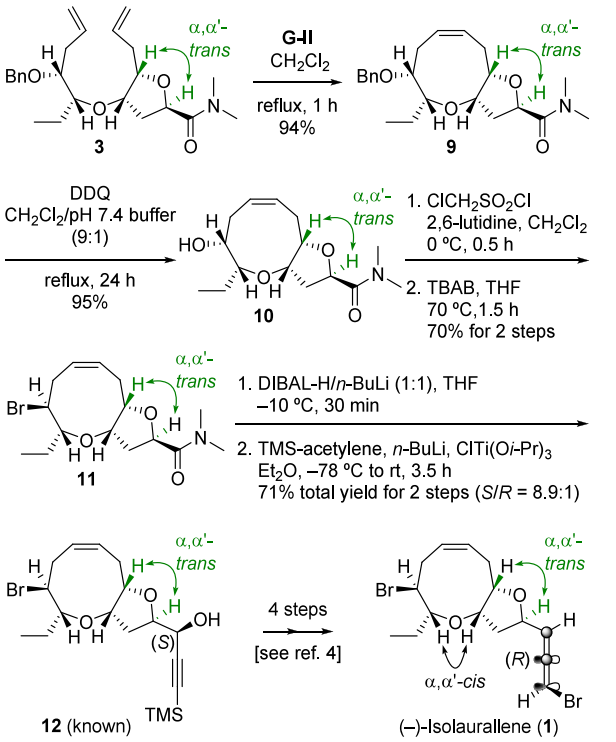
Scheme 3. Stereodivergent Intramolecular Amide Enolate Alkylation of 5



presumably through chelate-controlled transition state geometry **A**. On the other hand, we were delighted to find that subjecting **5** to KHMDS in THF at −78 to −50 °C for 2 h in the presence of 18-crown-6 delivered *α,α′-trans*-THF **3** in 90% yield as a single stereoisomer (by ^{1}H NMR analysis). The stereoselectivity observed in this case could be rationalized by invoking a nonchelated transition state geometry (**B1**), which would minimize the steric repulsions between the bulky alkoxy group (OR) and the amide enolate moiety that would exist in the alternative nonchelated transition state geometry (**B2**), as depicted in the scheme. It is noteworthy that our non-chelate-controlled IAEA protocol markedly improved the moderate stereoselectivity previously observed in the construction of *α,α′-trans*-THF **D** via INAA of **C** (*trans* only for IAEA vs 5.8:1 *trans*:*cis* for INAA).[4]

Having accomplished the highly stereoselective construction of the key intermediates *α,α′-trans*-THF amide **3** and *α,α′-cis*-THF amide **4** in a stereodivergent manner, we first turned our attention to the synthesis of (−)-isolaurallene (**1**) (Scheme 4). Thus, RCM of diene **3** using the Grubbs second-generation

Scheme 4. Formal Total Synthesis of (−)-Isolaurallene (1)



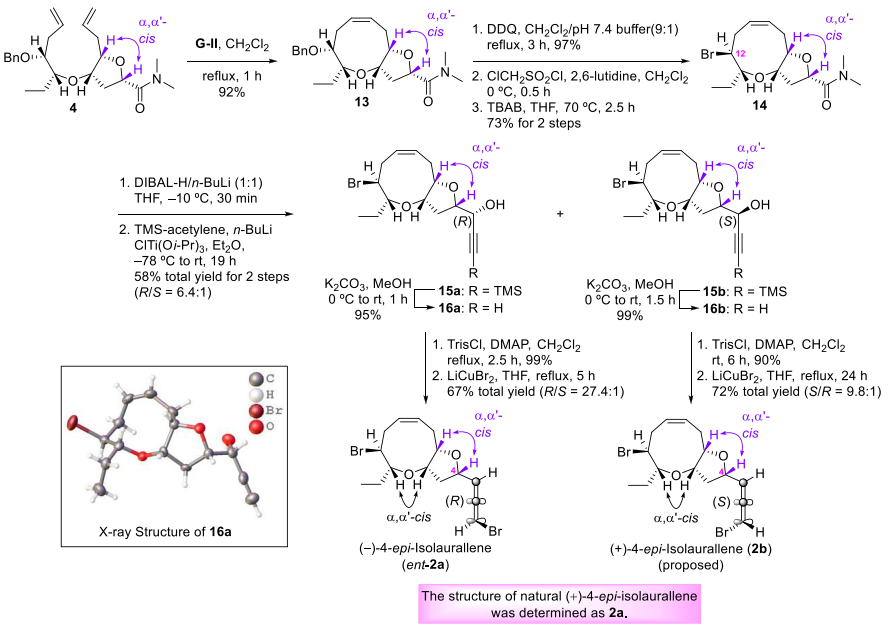
catalyst [**G-II**, $(H_2IMes)(Cy_3P)Cl_2Ru{=}CHPh$] produced bicyclic oxonene **9** in an excellent yield (94%). Removal of the benzyl group in **9** under the reaction conditions reported by Yonemitsu and co-workers[8] and subsequent bromination of the resulting secondary alcohol **10** with inversion of configuration using Nakata's two-step chloromesylate-mediated S_N2 displacement furnished bromide **11** (67% for three steps).[9] Finally, reduction of tetrahydrofuranyl *α*-alkoxyamide **11** using the ate complex[10] derived from DIBAL-H and *n*-BuLi followed by addition of titanium TMS-acetylide to the corresponding aldehyde under the conditions developed by Overman and co-workers[11] gave rise to known (*S*)-TMS-propargylic alcohol **12** (8.9:1 *S*:*R*, 71% total yield for two steps), which was previously transformed into (−)-isolaurallene (**1**) in four steps,[4] thus completing the formal total synthesis of **1**.

Next, we focused on the synthesis and structure determination of newly isolated (+)-4-*epi*-isolaurallene (**2**) (Scheme 5). Thus, oxonene **13**, which was obtained from *α,α′-cis*-THF **4** via RCM in excellent yield (92%), was converted into bromide **14** (71% for three steps; debenzylation, chloromesylation, and bromination in that order) under conditions analogous to those used for 12-bromo *α,α′-trans*-amide **11** in the synthesis of (−)-isolaurallene (**1**) (see Scheme 4).

With the C12 bromide functionality installed, the remaining task to complete the synthesis of *ent*-**2a** and **2b** was the elaboration of the (*R*)- and (*S*)-bromoallene units at C(4), which required propargylic alcohols **16a** and **16b**, respectively. By making use of the analogous approach employed in the preparation of TMS-propargylic alcohol **12** from *α,α′-trans*-THF amide **3** (Scheme 4), partial reduction of *α,α′-cis*-THF amide **14** with the DIBAL-H/*n*-BuLi ate complex followed by addition of titanium TMS-acetylide to the corresponding aldehyde gave rise to (*R*)- and (*S*)-TMS-propargylic alcohols **15a** and **15b**, respectively (58% total yield, 6.4:1 **15a**:**15b** ratio

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Scheme 5. Structure Determination of (+)-4-*epi*-Isolaurallene (2a) via the Total Synthesis of *ent*-2a and 2b



determined by ^{1}H NMR analysis), which were desilylated by being treated with K_2CO_3 in MeOH to afford **16a** and **16b** in 95% and 99% yields, respectively.[12,13] The structure of **16a** was confirmed by means of an X-ray crystallographic analysis (Scheme 5).

With propargylic alcohols **16a** and **16b** in hand, we proceeded to complete the total synthesis of *ent*-**2a** and **2b**, respectively. Thus, trisylation of propargylic alcohols **16a** and **16b** gave the corresponding trisylates (99% and 90% yields, respectively), which were subjected to a Cu-mediated *anti*-S_N2' bromination by exposure to $LiCuBr_2$ to deliver *ent*-**2a** and **2b** [67% (27.4:1 *R*:*S*) and 72% (9.8:1 *S*:*R*) total yields, respectively]. The ^{1}H and ^{13}C NMR spectra of our synthetic material **2b** differed from those of natural (+)-4-*epi*-isolaurallene. Conversely, the spectral characteristics and optical rotation of *ent*-**2a** were in good agreement with those reported for the natural product with the exception of the sign for the optical rotation, thereby confirming the structure of natural (+)-4-*epi*-isolaurallene as **2a** ($[\alpha]_D^{25}$ −36.4 (*c* 0.87, $CHCl_3$); natural form lit.[5] $[\alpha]_D^{25}$ +56.1 (*c* 0.46, $CHCl_3$)).

In conclusion, we accomplished the highly stereoselective construction of *α,α′-trans*-THF **3** and *α,α′-cis*-THF **4** from a common cyclization substrate **5** using a highly stereodivergent IAEA reaction in a non-chelate-controlled and chelate-controlled manner, respectively. Our non-chelate-controlled IAEA protocol markedly improved upon the moderate stereoselectivity previously observed in the construction of the *α,α′-trans*-THF ring via INAA (*trans* only via IAEA vs 5.8:1 *trans*:*cis* via INAA). By applying this methodology, the formal total synthesis of (−)-isolaurallene (**1**) was achieved and the structure of (+)-4-*epi*-isolaurallene (**2**) was confirmed unambiguously to be **2a** via the total synthesis of its enantiomer (*ent*-**2a**). Furthermore, our stereodivergent IAEA protocol could be applicable to stereoselective syntheses of 2,5-*cis*- and 2,5-*trans*-3-hydroxy tetrahydrofuranoid natural products, in particular, dioxabicyclic bromoallene *Laurencia* marine natural products with a 2,10-dioxabicyclo[7.3.0]dodecane.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c00480.

Experimental procedures, characterization data, and copies of NMR spectra (PDF)

Accession Codes

Deposition Number 2395537 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

[▽]Y.P. and H.J. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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(12) (*R*)-TMS-propargylic alcohol **15a** could be converted into (*S*)-propargylic alcohol **16b** by a two-step sequence: Mitsunobu inversion (58%) and saponification/desilylation (94%). See the Supporting Information for details.

(13) Application of the direct ketone synthesis protocol to α-alkoxy amide **14** with lithium TIPS-acetylide in the presence of $BF_3{\cdot}OEt_2$ (THF, −78 °C to rt, 2.5 h, 78%) followed by L-Selectride reduction (THF, −78 °C, 1.5 h) of the corresponding TIPS-ynone afforded (*R*)-TIPS-propargylic alcohol as the major isomer (73% total yield, 3:1 *R*:*S* determined by ^{1}H NMR analysis) predicted by an *anti*-Felkin−Ahn selectivity (see the Supporting Information for details).